

HW 1: Schrödinger Equation(s)

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Question 1) Compatible observables are those for which

$$[\hat{A}, \hat{B}] = 0, \quad (1)$$

and incompatible observables are those for which

$$[\hat{A}, \hat{B}] \neq 0. \quad (2)$$

You already know that position and momentum are incompatible observables, since

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}\mathbf{1} \text{ for the individual } i = x, y, z \text{ components.} \quad (3)$$

- Prove that a Hermitian operator has i) real eigenvalues *and* ii) eigenvectors associated with different eigenvalues are orthogonal.
- Prove that the eigenstates of $\hat{A}$ are also eigenstates of $\hat{B}$ if $\hat{A}$ and $\hat{B}$ are compatible observables. You can assume non-degenerate eigenvalues for $\hat{A}$. When the eigenbasis of $\hat{A}$ forms a complete basis, this proves that compatible observables can be simultaneously diagonalized.
- Show that if $\hat{A}$ and $\hat{B}$ are compatible observables and $\hat{A}$ and $\hat{C}$ are compatible observables, then $\hat{B}$ and $\hat{C}$ are also compatible observables, assuming non-degenerate eigenvalues for $\hat{A}$.
- As a special case of the above, show that if $\hat{A}$ and $\hat{B}$ are compatible observables, then any function of $\hat{A}$, say $f(\hat{A})$, is also compatible with $\hat{B}$. Assume non-degenerate eigenvalues for $\hat{A}$, $f(\hat{A})|a\rangle = f(a)|a\rangle$, and that $f(\hat{A})$ is Hermitian.

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Question 2) The time-dependent Schrödinger equation in operator, basis-independent form is

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$. As we will show later, the momentum operator $\hat{\mathbf{p}}$ acts on position wavefunctions as $-i\hbar\nabla$. To be more precise, we have

$$\langle \mathbf{x} | \hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{x}} \langle \mathbf{x} | \text{ and } \hat{\mathbf{p}} | \mathbf{x} \rangle = +i\hbar \nabla_{\mathbf{x}} | \mathbf{x} \rangle.$$

A key manipulation is how to apply the momentum operator on a state $|\psi\rangle$ in the position representation:

$$\begin{aligned} \langle \mathbf{x} | \hat{\mathbf{p}} | \psi \rangle &= -i\hbar \nabla_{\mathbf{x}} \langle \mathbf{x} | \psi \rangle = -i\hbar \nabla_{\mathbf{x}} \psi(\mathbf{x}), \\ \langle \psi | \hat{\mathbf{p}} | \mathbf{x} \rangle &= +i\hbar \nabla_{\mathbf{x}} \langle \psi | \mathbf{x} \rangle = +i\hbar \nabla_{\mathbf{x}} \psi^*(\mathbf{x}). \end{aligned} \quad (4)$$

Note that these are complex conjugates of each other as it must be for a Hermitian operator.

Note: $\nabla_{\mathbf{x}}$ is differentiating with respect to $\mathbf{x}$. Note also the minus signs here.

- derive the time-dependent Schrödinger equation for the energy wavefunction $\psi_n(t) = \langle n | \psi(t) \rangle$ for a discrete spectrum, and then again for $\psi(E, t) = \langle E | \psi(t) \rangle$ in a continuous

spectrum. Note that $\hat{H}(t) |n\rangle = E_n |n\rangle$ for a discrete spectrum and $\hat{H} |E\rangle = E |E\rangle$ for a continuous spectrum.

By spectrum I mean the set of all possible energy eigenvalues of the Hamiltonian operator. Note that in this language, E_n and $|n\rangle$ are time-independent.

- b) Derive the time-dependent Schrödinger equation in position space (in 3 spatial dimensions), i.e., find the differential equation obeyed by the wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$.

Use a Hamiltonian of the form $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$ and a potential that depends only on position, that is, $V(\hat{\mathbf{x}}) |\mathbf{x}\rangle = V(\mathbf{x}) |\mathbf{x}\rangle$ with $V(\mathbf{x})$ being just a number for each position vector $\mathbf{x}$.



Bonus) Repeat the same steps to derive the time-dependent Schrödinger equation in momentum space, i.e., find the *integro-differential* equation obeyed by $\phi(\mathbf{p}, t) = \langle \mathbf{p} | \psi(t) \rangle$.

The momentum and position kets are related by the Fourier transform, $\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$.

Your answer should be an integro-differential equation that depends on the Fourier transform of the potential:

$$\langle \mathbf{p} | \hat{V} | \mathbf{p}' \rangle = \tilde{V}(\mathbf{p} - \mathbf{p}') = \frac{1}{(2\pi\hbar)^3} \int d^3\mathbf{x} e^{-i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} V(\mathbf{x}).$$

Some useful relations:

$$\int d^3\mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x}| = \hat{I}, \quad \int d^3\mathbf{p} |\mathbf{p}\rangle \langle \mathbf{p}| = \hat{I}, \quad (5)$$

$$\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \text{ and its hermitian conjugate version} \quad (6)$$

$$(7)$$

$$\int d^3\mathbf{x} \langle \mathbf{p} | \mathbf{x} \rangle \langle \mathbf{x} | \mathbf{p}' \rangle = \int d^3\mathbf{x} \frac{1}{(2\pi\hbar)^3} e^{-i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} = \delta(\mathbf{p} - \mathbf{p}') \quad (8)$$

$$\int d^3\mathbf{p} \langle \mathbf{x} | \mathbf{p} \rangle \langle \mathbf{p} | \mathbf{x}' \rangle = \int d^3\mathbf{p} \frac{1}{(2\pi\hbar)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{x}')/\hbar} = \delta(\mathbf{x} - \mathbf{x}') \quad (9)$$



Question 1.0: Some simple proofs to get us used to the notation.

a) Take $A^\dagger = A$ or equivalently,

$$\langle \alpha | A | \beta \rangle = \langle \beta | A | \alpha \rangle^*. \quad (10)$$

Take its eigenvectors $|a\rangle$ and eigenvalues a such that $A|a\rangle = a|a\rangle$. Then,

$$\langle a | A | a \rangle = a \langle a | a \rangle = a \quad \text{and} \quad \langle a | A | a \rangle = \langle a | A | a \rangle^* \implies a = a^*. \quad (11)$$

Equivalently, do

$$\langle a' | A | a \rangle = a'^* \langle a' | a \rangle, \quad \langle a' | A | a \rangle = a \langle a' | a \rangle. \quad (12)$$

where we acted with A on the left on the left equation, and on the right on the right equation. Subtract the right equation from the left one, we get

$$(a' - a^*) \langle a' | a \rangle = 0. \quad (13)$$

If $a = a'$, then we reach the same conclusion as above. If $a \neq a'$, then $\langle a' | a \rangle \stackrel{!}{=} 0$, which shows that eigenvectors corresponding to distinct eigenvalues are orthogonal. Note this says nothing about the orthogonality potential degenerate eigenvectors.

b) Taking $[A, B] = 0$ and acting with it on an eigenvector $|a\rangle$ of A :

$$[A, B] |a\rangle = 0 \implies AB|a\rangle = BA|a\rangle = B(A|a\rangle) = B(a|a\rangle) = aB|a\rangle. \quad (14)$$

Because the eigenspace of A is non-degenerate, there can only be one eigenvector for each eigenvalue, which implies that $B|a\rangle$ must be proportional to $|a\rangle$ itself. Therefore, $|a\rangle$ is also an eigenvector of B .

If the eigenspace of A is degenerate, then $B|a\rangle$ could be a linear combination of all eigenvectors of A with the same eigenvalue a , and we cannot conclude that $|a\rangle$ is an eigenvector of B in that case.



If A and B commute and have a common set of eigenvectors, then we can simultaneously diagonalize them. This means that there exists a basis in which both A and B are diagonal, and their eigenvectors form a complete orthonormal set. This is trivially the case for many operators. For example,

$$\hat{x}, \hat{y}, \hat{z} \quad (15)$$

all commute with each other and therefore

$$|x, y, z\rangle \quad (16)$$

provides a diagonal basis for all. Another trivial example is $\hat{x}$ and $\hat{x}^2$. They obviously commute and in this case $|x\rangle$ provides a diagonal basis for both.

c) If $[A, B] = 0$ and $[A, C] = 0$, then

$$B|a\rangle = b|a\rangle, \quad C|a\rangle = c|a\rangle \quad (17)$$

and

$$[B, C] |a\rangle = (BC - CB) |a\rangle = B(c|a\rangle) - C(b|a\rangle) = cB|a\rangle - bC|a\rangle = cb|a\rangle - bc|a\rangle = 0 \quad (18)$$

Note this relied on the non-degeneracy of the eigenspace of A that we used in part b).

d) This is a special case of part c). Just make $\hat{C} = f(\hat{A})$ and note that $[A, f(\hat{A})] = 0$, which follows from just Taylor expanding $f(\hat{A})$ in powers of $\hat{A}$ and noting that $[\hat{A}, \hat{A}^n] = 0$ for any power of n .



Question 2.0: This problem is all about letting the formalism do the work and gives us a glimpse of how powerful the basis-independent Schrödinger equation is.

a) Both the discrete and the continuous cases lead to a very similar equation, we just have to be careful with the meaning of the different objects. Start with the discrete case

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \quad (19)$$

contracting both sides with $\langle n|$ gives

$$i\hbar \frac{d}{dt} \langle n|\psi(t)\rangle = \langle n|\hat{H}|\psi(t)\rangle \implies i\hbar \frac{d}{dt} \psi_n(t) = E_n \psi_n(t), \quad (20)$$

where we defined $\psi_n(t) = \langle n|\psi(t)\rangle$ and used the complex conjugated version of $\hat{H}|n\rangle = E_n|n\rangle$. Note that we are working with fixed energy/position/momentum basis here, so the kets (and also the bras) are time-independent. Once we place the bra inside the time derivative, it then becomes a partial derivative since the wavefunction depends now on both time and space.

For the continuous case, we do the same, but dot the first equation with $\langle E|$ instead to get $\psi(E, t) = \langle E|\psi(t)\rangle$.



Can you repeat these steps without assuming the hermiticity of the Hamiltonian?

b) We want to express everything in terms of the position-space wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x}|\psi(t)\rangle$. To that end, we will contract both sides of the operator Schrödinger equation with $\langle \mathbf{x}|$ from the left (i.e., apply the linear operator $\langle \mathbf{x}|$ to both sides):

$$i\hbar \frac{\partial}{\partial t} \langle \mathbf{x}|\psi(t)\rangle = \langle \mathbf{x}|\hat{H}|\psi(t)\rangle \quad (21)$$

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \langle \mathbf{x}|\left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})\right)|\psi(t)\rangle. \quad (22)$$

where we just used the definition of the position-space wavefunction.

The RHS is more interesting. Let us insert the completeness relation of the position basis right before the ket $|\psi(t)\rangle$:

$$\int d^3\mathbf{x}' \langle \mathbf{x}|\left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})\right)|\mathbf{x}'\rangle \langle \mathbf{x}'|\psi(t)\rangle = \int d^3\mathbf{x}' \langle \mathbf{x}|\left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})\right)|\mathbf{x}'\rangle \psi(\mathbf{x}', t).$$

Curiously, our RHS became a sort of expectation value times the wavefunction. Let us find what this expectation value is. First, consider the potential term. This one is assumed to be simple enough that $V(\hat{\mathbf{x}})|\mathbf{x}'\rangle = V(\mathbf{x}')|\mathbf{x}'\rangle$. Now, moving onto the kinetic term, we can use the fact the momentum operator rules provided in the question. Note that acting with $\hat{\mathbf{p}}$ on a state ought to return a 3D vector times a state. Applying it twice, $\hat{\mathbf{p}}^2$, ought to return a scalar. Being very deliberate about each step, here's how you would do it:

$$\langle \mathbf{x} | \hat{\mathbf{p}}^2 | \mathbf{x}' \rangle = \langle \mathbf{x} | \hat{\mathbf{p}} \cdot \hat{\mathbf{p}} | \mathbf{x}' \rangle = (-i\hbar \vec{\nabla}_{\mathbf{x}}) \cdot \langle \mathbf{x} | \hat{\mathbf{p}} | \mathbf{x}' \rangle = -\hbar^2 (\nabla_{\mathbf{x}}^2) \langle \mathbf{x} | \mathbf{x}' \rangle = -\hbar^2 (\nabla_{\mathbf{x}}^2) \delta(\mathbf{x} - \mathbf{x}'). \quad (23)$$

Under the integral sign over $\mathbf{x}'$, the Laplacian $\nabla_{\mathbf{x}}^2 \delta(\mathbf{x} - \mathbf{x}')$ can be replaced by $\nabla_{\mathbf{x}'}^2 \delta(\mathbf{x} - \mathbf{x}')$.

Putting this all together, we have

$$\langle \mathbf{x} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{x}' \rangle = \langle \mathbf{x} | \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}'}^2 + V(\mathbf{x}') \right) | \mathbf{x}' \rangle \quad (24)$$

$$= \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}'}^2 + V(\mathbf{x}') \right) \langle \mathbf{x} | \mathbf{x}' \rangle \quad (25)$$

$$= \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}'}^2 + V(\mathbf{x}') \right) \delta(\mathbf{x} - \mathbf{x}'), \quad (26)$$

where we used the fact that $\langle \mathbf{x} | \mathbf{x}' \rangle = \delta(\mathbf{x} - \mathbf{x}')$. Under the integral, we can now perform the integration over $\mathbf{x}'$ using the properties of the Dirac delta to find

$$\langle \mathbf{x} | \hat{H} | \psi(t) \rangle = \int d^3 \mathbf{x}' \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}'}^2 + V(\mathbf{x}') \right) \delta(\mathbf{x} - \mathbf{x}') \psi(\mathbf{x}', t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t). \quad (27)$$

The desired result follows:

$$\boxed{i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t).} \quad (28)$$

Bonus) We will repeat the same steps, but now using the momentum basis. Contracting both sides of the operator Schrödinger equation with $\langle \mathbf{p} |$ from the left, we have

$$i\hbar \frac{\partial}{\partial t} \langle \mathbf{p} | \psi(t) \rangle = \langle \mathbf{p} | \hat{H} | \psi(t) \rangle \quad (29)$$

$$i\hbar \frac{\partial}{\partial t} \phi(\mathbf{p}, t) = \langle \mathbf{p} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \psi(t) \rangle. \quad (30)$$

Let us insert the completeness relation of the momentum basis right before the ket $|\psi(t)\rangle$ to get the RHS of Eq. (30):

$$\text{RHS} = \int d^3 \mathbf{p}' \langle \mathbf{p} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{p}' \rangle \langle \mathbf{p}' | \psi(t) \rangle = \int d^3 \mathbf{p}' \langle \mathbf{p} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{p}' \rangle \phi(\mathbf{p}', t).$$

Now, it is the kinetic term that becomes trivial, since $\hat{\mathbf{p}} |\mathbf{p}'\rangle = \mathbf{p}' |\mathbf{p}'\rangle$. The potential term, on the other hand, is trickier now since we do not know what form it takes in momentum space. This is not particularly surprising since the whole point of working in momentum space should be that we know how to write down the problem in momentum space, but we are given a potential in position space. Let us fix this by inserting the completeness relation of the position basis right after the potential operator:

$$\langle \mathbf{p} | V(\hat{\mathbf{x}}) | \mathbf{p}' \rangle = \int d^3 \mathbf{x}' \langle \mathbf{p} | V(\hat{\mathbf{x}}) | \mathbf{x}' \rangle \langle \mathbf{x}' | \mathbf{p}' \rangle \quad (31)$$

$$= \int d^3 \mathbf{x}' V(\mathbf{x}') \langle \mathbf{p} | \mathbf{x}' \rangle \langle \mathbf{x}' | \mathbf{p}' \rangle. \quad (32)$$

If you recall, the momentum and position kets are related by the Fourier transform,¹

$$\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}.$$

Therefore, we have that the sandwich above is just the Fourier transform of the potential(!):

$$\langle \mathbf{p} | V(\hat{\mathbf{x}}) | \mathbf{p}' \rangle = \int d^3\mathbf{x}' V(\mathbf{x}') \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}\cdot\mathbf{x}'/\hbar} \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}'\cdot\mathbf{x}'/\hbar} \quad (33)$$

$$= \frac{1}{(2\pi\hbar)^3} \int d^3\mathbf{x}' V(\mathbf{x}') e^{i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{x}'/\hbar} \equiv \tilde{V}(\mathbf{p}' - \mathbf{p}), \quad (34)$$

where we made it apparent that $\tilde{V}(\mathbf{q})$ is simply the Fourier transform of the potential $V(\mathbf{x})$.

Putting this all together,

$$\langle \mathbf{p} | \hat{H} | \psi(t) \rangle = \int d^3\mathbf{p}' \left(\frac{\mathbf{p}'^2}{2m} + \tilde{V}(\mathbf{p}' - \mathbf{p}) \right) \phi(\mathbf{p}', t). \quad (35)$$

and

$$\boxed{i\hbar \frac{\partial}{\partial t} \phi(\mathbf{p}, t) = \frac{\mathbf{p}^2}{2m} \phi(\mathbf{p}, t) + \int d^3\mathbf{p}' \tilde{V}(\mathbf{p}' - \mathbf{p}) \phi(\mathbf{p}', t).} \quad (36)$$

Bonus² Let's see how equations Eqs. (28) and (36) are equivalent by Fourier transforming the position-space Schrödinger equation:

$$\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle = \int d^3\mathbf{p} \langle \mathbf{x} | \mathbf{p} \rangle \phi(\mathbf{p}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \quad (37)$$

Starting from

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t),$$

we will substitute the Fourier transform relation for $\psi(\mathbf{x}, t)$:

$$i\hbar \frac{\partial}{\partial t} \left(\frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \right) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \left(\frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \right).$$

Next, we will multiply both sides by $\frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}$ and integrate over all space $d^3\mathbf{x}$:

$$i\hbar \frac{\partial}{\partial t} \int d^3\mathbf{x} \frac{1}{(2\pi\hbar)^3} \int d^3\mathbf{p} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \quad (38)$$

$$= \int d^3\mathbf{x} \frac{e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3} \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t). \quad (39)$$

The LHS is straightforward, since the integral over $\mathbf{x}$ just gives us a Dirac delta:

$$\text{LHS} = i\hbar \frac{\partial}{\partial t} \int d^3\mathbf{p} \delta(\mathbf{p} - \mathbf{p}') \phi(\mathbf{p}, t) = i\hbar \frac{\partial}{\partial t} \phi(\mathbf{p}', t). \quad (40)$$

The RHS is more involved. Let us split it into the kinetic and potential terms. The kinetic term is (noting that ∇ acts on $\mathbf{x}$ and the exponential we inserted on both sides is not acted

¹See Sec. 1.7.3 in Sakurai and Napolitano. The normalization constant is fixed by the normalization condition $\langle \mathbf{p} | \mathbf{p}' \rangle = \delta(\mathbf{p} - \mathbf{p}')$.

upon by the derivative):

$$\text{RHS}_{\text{kin}} = -\frac{\hbar^2}{2m} \int d^3\mathbf{x} \frac{e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3} \nabla^2 \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \quad (41)$$

$$= -\frac{\hbar^2}{2m} \int d^3\mathbf{p} \phi(\mathbf{p}, t) \int d^3\mathbf{x} \frac{e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3} \nabla^2 e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \quad (42)$$

$$= -\frac{\hbar^2}{2m} \int d^3\mathbf{p} \phi(\mathbf{p}, t) \int d^3\mathbf{x} \frac{e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3} \left(-\frac{\mathbf{p}^2}{\hbar^2} \right) \quad (43)$$

$$= \int d^3\mathbf{p} \phi(\mathbf{p}, t) \frac{\mathbf{p}^2}{2m} \delta(\mathbf{p} - \mathbf{p}') = \frac{\mathbf{p}'^2}{2m} \phi(\mathbf{p}', t). \quad (44)$$

The potential term is

$$\text{RHS}_{\text{pot}} = \int d^3\mathbf{x} \frac{1}{(2\pi\hbar)^3} V(\mathbf{x}) \int d^3\mathbf{p} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \quad (45)$$

$$= \int d^3\mathbf{p} \phi(\mathbf{p}, t) \int d^3\mathbf{x} \frac{1}{(2\pi\hbar)^3} V(\mathbf{x}) e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \quad (46)$$

$$= \int d^3\mathbf{p} \phi(\mathbf{p}, t) \tilde{V}(\mathbf{p} - \mathbf{p}'), \quad (47)$$

where we recognized the Fourier transform of the potential once again. Putting this all together, we have

$$i\hbar \frac{\partial}{\partial t} \phi(\mathbf{p}', t) = \left(\frac{\mathbf{p}'^2}{2m} + \int d^3\mathbf{p} \tilde{V}(\mathbf{p} - \mathbf{p}') \phi(\mathbf{p}, t) \right),$$

which is exactly the momentum-space Schrödinger equation we derived in part (b), thus proving the equivalence of the two formulations.